

MITRE ATT&CK Mapping

MITRE ATT&CK Framework Correlation

The investigation findings were mapped against the MITRE ATT&CK framework to classify attacker tactics and techniques observed throughout the compromise lifecycle. Mapping detected activity to MITRE ATT&CK techniques improves incident classification, enhances threat intelligence correlation, and supports standardized Security Operations Center (SOC) investigation workflows.

Investigation Activity	MITRE ATT&CK Technique	Technique ID	Description
Reverse shell execution using Socat	Command and Scripting Interpreter: Unix Shell	T1059.004	The attacker used shell execution through Socat to establish remote command execution access on the compromised Linux system.
Outbound reverse shell TCP connection	Application Layer Protocol	T1071	The reverse shell communicated through outbound TCP traffic to an external attacker-controlled IP address.
Reverse shell process execution	System Services	T1569	The attacker executed malicious processes and shell commands through the compromised application environment.
Process tree tracing and shell execution	Process Discovery	T1057	Process tracing techniques revealed attacker process relationships and execution chains within the compromised environment.

Privilege escalation using su root	Abuse Elevation Control Mechanism	T1548	The attacker successfully escalated privileges by obtaining root-level access through the su command.
Root-level shell execution	Valid Accounts	T1078	Elevated root privileges enabled privileged command execution and administrative-level access within the system.
Cron persistence modification	Scheduled Task/Job: Cron	T1053.003	The attacker used cron job modification techniques to establish persistence within the Linux environment.
SSH authorized key persistence	Account Manipulation	T1098	Unauthorized modification of SSH authorized keys allowed persistent passwordless remote access.
Systemd service persistence	Create or Modify System Process	T1543	Malicious systemd service files were created to establish persistence and execute attacker-controlled services.
Malicious service deployment using wget	Ingress Tool Transfer	T1105	External tools and malicious service configurations were downloaded into the compromised environment.
Suspicious shell execution activity	Command and Scripting Interpreter	T1059	Multiple shell commands were executed throughout the compromise lifecycle for attacker operations and persistence.
Elevated root activity correlation	Privilege Escalation	T1068	Root-level command execution confirmed

			extensive privileged attacker activity after successful escalation.
Outbound network communication correlation	Exfiltration Over C2 Channel	T1041	Suspicious outbound communication behavior was observed during attacker command-and-control activity.

Key MITRE ATT&CK Findings

- Multiple attacker tactics were identified across execution, privilege escalation, persistence, and command-and-control phases.
- Reverse shell behavior aligned closely with Unix shell execution and outbound command-and-control communication techniques.
- Persistence mechanisms involved cron jobs, SSH authorized key modification, and malicious systemd service creation.
- Privileged root activity demonstrated successful escalation and extended attacker control over the compromised Linux environment.
- MITRE ATT&CK mapping improved visibility into attacker objectives, post-exploitation behavior, and persistence techniques observed during the investigation.